# Noah Paoa, Ph.D.

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## PROFESSIONAL SUMMARY

Geospatial scientist and PhD graduate specializing in coastal hazards, sea-level rise modeling, and remote sensing. Experienced in geodetic surveying, LiDAR processing, UAV mapping, and large-scale geospatial data analysis using Python and GIS platforms. Proven ability to generate and condition digital elevation models, develop probability-based flood models, and translate complex geospatial data into intuitive visualizations and actionable insights for coastal management, hazard mitigation and environmental planning.

## TECHNICAL SKILLS

**Geospatial & Remote Sensing**
ArcGIS Pro, QGIS, Global Mapper, GMT
LiDAR processing (LAStools, CloudCompare),
DEM production

**Programming & Data Analysis**
Python (NumPy, Pandas, Geopandas, GDAL, ArcPy)
Geospatial pipelines, data processing automation

**Field and Data Acquisition**
GNSS, UAV mapping, bathymetry, instrumentation deployment, geodetic surveying

**Modeling & simulation**
XBeach, SFINCS
Probability-based flood modeling
Coastal hazard assessment

## PROFESSIONAL EXPERIENCE

**Graduate Research Assistant**

University of Hawaiʻi at Mānoa-Honolulu, HI | 2020–2026

- Coordinated and conducted geospatial field surveys in remote field environments.
- Deployed and managed coastal monitoring instrumentation in remote field environments (pressure sensors, GNSS receivers, echosounders).
- Processed and integrated GNSS, UAV, LiDAR, and bathymetric datasets to generate high-resolution DEMs for coastal hazard modeling.
- Built automated Python pipelines for geospatial data processing and visualization, reducing manual processing time and enabling scalable analysis workflows.
- Performed 1D/2D coastal flood simulations using XBeach to evaluate wave-driven flooding scenarios.
- Enhanced and implemented probability-based sea-level rise flood models using Python, expanding analytical capabilities and generating additional hazard scenarios from existing datasets.

**Undergraduate Researcher**
University of Oregon — Eugene, OR | 2017–2018

- Quantified erosion rates using field instrumentation and time-series image analysis in badland environments.
- Developed scripts to detect and analyze surface cracking patterns from imagery datasets.
- Supported LiDAR-based landslide mapping and volumetric analysis in the Cascade Range.
- Contributed to post-fire sediment transport studies through field reconnaissance and data analysis.

## PUBLICATIONS


**Paoa, N.**, Fletcher, C. H., Quintus, S., Barbee, M., Anderson, T. R., Habel, S., Kane, H. Multi-hazard assessment of sea-level rise impacts on burial sites of Oʻahu, Hawaiʻi. (in prep.)

Habel, S., Fletcher, C. H., Andrade, H., Azouri, A., Anderson, T. R., Barbee, M., Budge, J., Fornace, K. L., Kapono-Dye, A., Kimura, S., Lee, C., Meguro, W., Mikkelsen, A. B., Moskvichev, R., Nicolow, J., Obara, C., **Paoa, N.**, Stilgenbauer, J., Tavares, K., Yamamoto, K. Operationalizing Sea-Level Rise Modeling as a Governance Tool: Integrating Coastal Hazard Science into Adaptation Planning in Waikiki. (in prep.)

Mikkelsen, A. B., Fletcher, C. H., Moskvichev, R. U. 1, Habel, S., Kapono, A., Anderson, T. R., **Paoa, N**, Azouri, A., Mihami, F., Nicolow, J. C. Escalating infrastructure exposure to sea-level rise, wave-driven flooding, and coastal erosion on Oʻahu. (in prep.)

**Paoa, N.**, Fletcher, C. H., Barbee, M., Anderson, T. R., Habel, S., Wilkins Riroroko, G., Pakarati Trengove, S. (2026). Hydrostatic sea-level rise inundation impacts on *ahu* and harbors of Rapa Nui (Easter Island). *Sci Rep* **16**, 14509. https://doi.org/10.1038/s41598-026-45195-9

**Paoa, N.**, Fletcher, C. H., Azouri, A., Barbee, M., Anderson, T. R., Guiles, M., Habel, S., Thompson, P., McDonald, K., Tognacchini, C., Wilkins Riroroko, G., Luther, D. S (2025). Impacts of sea-level rise and wave inundation in the Tongariki Complex, Rapa Nui. Journal of Cultural Heritage, 75. https://doi.org/10.1016/j.culher.2025.07.004

**Paoa, N.**, Fletcher, C. H., Anderson, T. R., Coffman, M., & Habel, S. (2023). Probabilistic sea level rise flood projections using a localized ocean reference surface. Scientific Reports, 13(1). https://doi.org/10.1038/s41598-023-29297-2

## PROJECTS

- **Hawaiʻi Statewide Probabilistic Sea-Level Rise Flood Modeling**
  Developed probability-based hydrostatic sea-level rise flood extent layers for the state of Hawaiʻi using Python and geospatial datasets, supporting coastal hazard assessment and climate adaptation planning. (link)
- **Topobathymetric DEM Development – Rapa Nui (Easter Island)**
  Integrated multi-source elevation datasets (UAV photogrammetry, GNSS survey, and echosounder bathymetry) to develop the first high-resolution topo-bathymetric DEM of Rapa Nui, enabling improved coastal flood modeling. (link)

## EDUCATION

**PhD, Earth Sciences**
University of Hawaiʻi at Mānoa-Honolulu, HI | 2026 | GPA: 3.92

**MS, Earth Sciences**
University of Hawaiʻi at Mānoa-Honolulu, HI | 2021 | GPA: 3.90

**BS, Earth Sciences (Summa Cum Laude, Honors)**
University of Oregon-Eugene, OR | 2018 | GPA 3.90

## CERTIFICATIONS

- Deltares SFINCS Course (2025)
- Deltares XBeach Training (2022)
- Global Mapper LiDAR Training (2022)
- Global Mapper GIS Software Training (2021)

## ADDITIONAL INFORMATION

**Field & Instrumentation Experience**
GNSS (Emlid), pressure sensors (Solinst), echosounders, UAV systems

**Languages**
Spanish (Native), Rapa Nui (Native), English (Professional), Hawaiian (Basic)